TEMPERATURE CONTROLLED FAN

A Technical Report

Submitted By
Adib R Rafio

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Objective: A temperature controlled fan circuit.

Components:

- U741 op-amp
- NPN transistor
- Potentiometer
- NTC Thermistor
- Voltage controlled fan
- Diode
- Resistors
- Source

Working Principle:

The circuit uses a thermistor-based voltage divider to generate a temperature-dependent sensing voltage. This voltage is compared with an adjustable reference voltage using a potentiometer with an operational amplifier used as a comparator.

When the sensing voltage crosses the reference threshold, the comparator output changes to ON. This output drives the base of an NPN transistor through a current limiting resistor. The transistor acts as an switch, controlling the power supply to the fan.

A flyback diode is connected across the fan motor to suppress the inductive voltage generated when the transistor switches off. This helps protect the switching circuit from voltage spikes.

The potentiometer adjusts the reference threshold, allowing the switching point to be changed. The fan therefore switches on or off according to the relationship between the sensing and reference voltages.

Calculations:

Let the potentiometer be at x ohm for the calculation.
Upper resistance = (10k-x) ohms
Lower resistance= x ohms
Supply 9V
So the Reference voltage is at,

$$V_{ref} = 9 * \frac{x}{10k - x + x} \text{ V}$$

$$V_{ref} = \frac{9x}{10k} \text{V}$$

Let thermistor resistance be R_T. Applying Voltage divider on as threshold voltage = reference voltage,

$$\frac{9x}{10k} = 9 * \frac{Rt}{4.7k + RT}$$

$$\text{So, } R_T = \frac{4700x}{10000 - x} \text{ohms}$$

Result:

For various values of pot value, we may get different Reference Voltages & threshold Voltage.

Lower potentiometer resistance, x	Reference voltage, V_{ref}	Threshold thermistor resistance V_T
2k	1.8	1.175k
3k	2.7	2.014k
5k	4.5	4.7k
7k	6.3	10.97k
8k	7.2	18.8k

V_{ref} connects to the non-inverting input , and V_{sense} connects to the inverting input . So-

- At **high temperature**, R_T is below the threshold, $V_{sense} < V_{ref}$, so the comparator output is expected to go high and switch the transistor and fan on.
- At **low temperature**, R_T is above the threshold, $V_{sense} > V_{ref}$, so the comparator output is expected to go low and switch the fan off.

Circuit:

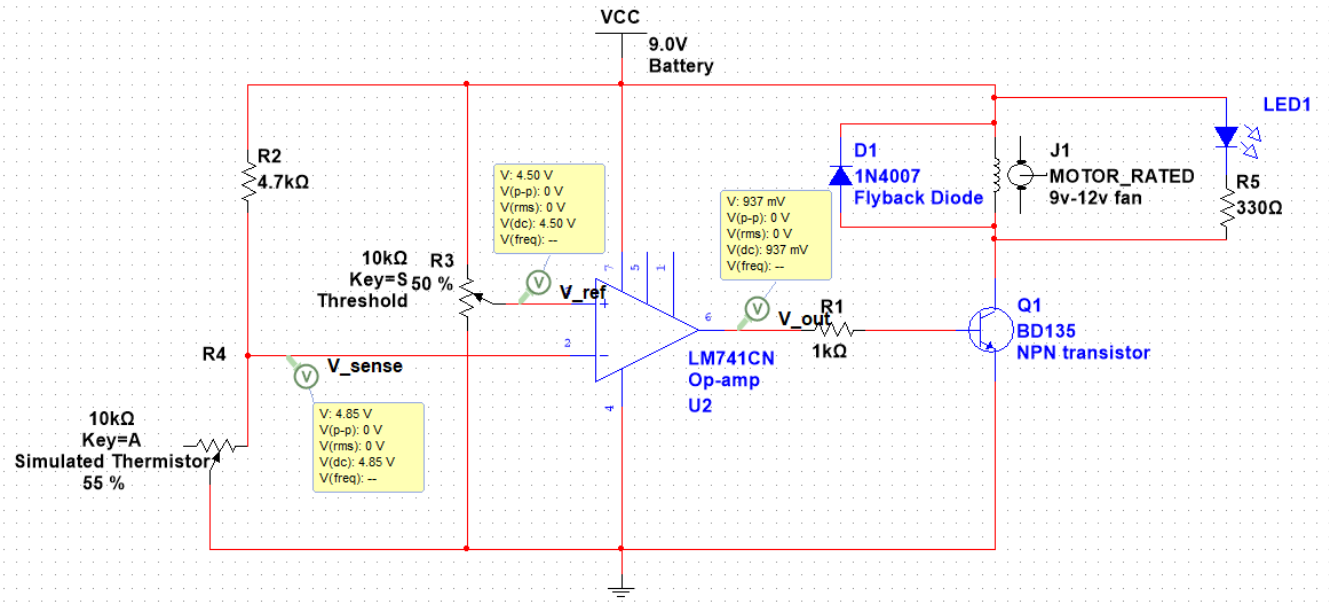


fig: LowTemperature State ($V_{sense} > V_{ref}$)

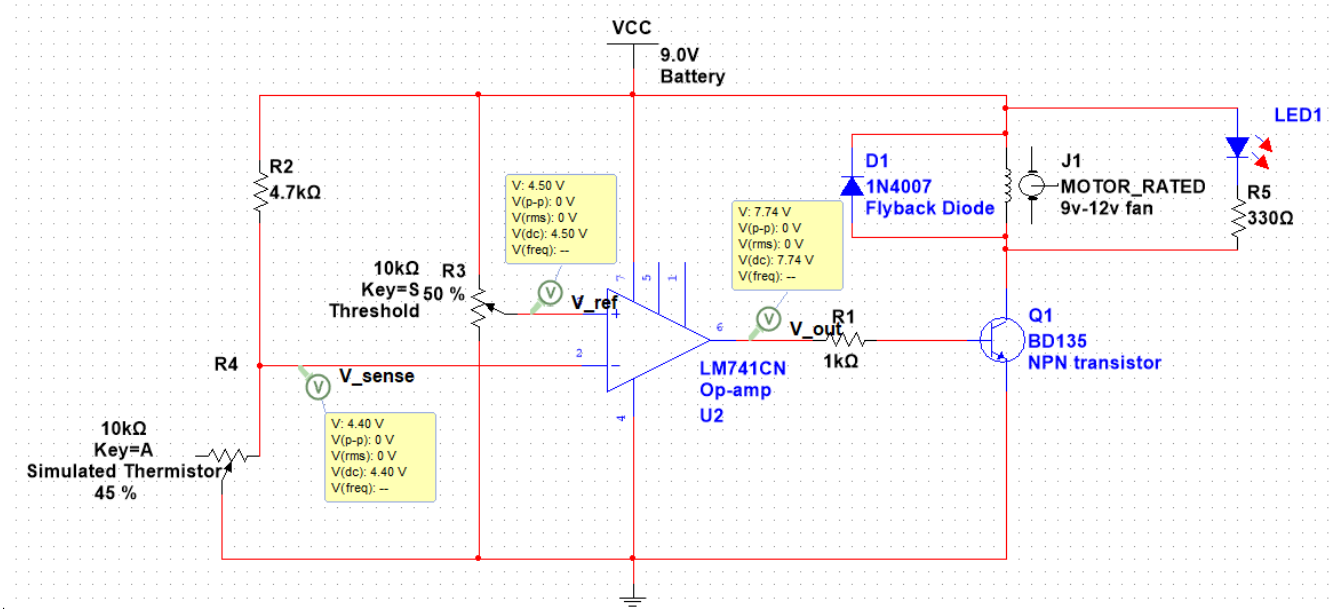


fig: High Temperature State ($V_{sense} < V_{ref}$)

Conclusion:

When temperature is low, the R_T is high. Thus, the reference voltage is high. $V_{sense} > V_{ref}$. So fan turns off.

Similarly, When temperature is high, the R_T is low. Thus, the reference voltage is low. $V_{sense} < V_{ref}$. So fan turns on.